National University of Computer and Emerging Sciences, Lahore Campus

Course:	Applied Calculus	Course Code:	MT-1001
Program:	BS(EE/RO)	Semester:	Fall 2022
Duration:	90 minutes	Total Marks:	50
Worksheet Date:	06-12-2022	Weight	6%
Section:	EE/RO	Page(s):	
Exam:	Full Book (Work Sheet)	Roll No:	
Name:			

Question#1: [CLO-05] [30 marks]

(a). Find the distance between the given parallel planes.

$$6z = 4y - 2x \quad ; \quad 9z = 1 - 3x + 6y$$

(b). Determine whether the planes are parallel, perpendicular, or neither. If neither, find the angle between them. (Round to one decimal place.)

$$5x + 2y - 3z = 2, \qquad y = 4x - 6z$$

(c). Find an equation of the plane.

The plane that passes through the point $(3, 1, 4)$ and contains the line of intersection of the planes

$$x + 2y + 3z = 1 \quad \text{and} \quad 2x - y + z = -3$$

Question #2: [CLO-04] Evaluate the integral. [20 marks]

Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

(a) $\int_1^{\infty} \frac{dx}{\sqrt{x}+x\sqrt{x}}$ (b) $\int_{-2}^{3} \frac{1}{x^4}\,dx$

Question #3: [CLO-01] [10 marks]

As dry air moves upward, it expands and in so doing cools at a rate of about $1^{o}\,C$ for $100 - m$ rise, up to about 12 km.

(a) If the ground temperature is 20°C, write a formula for the temperature at height h.

(b) What range of temperature can be expected if a plane takes off and reaches a maximum height of 5 km?

Question #4: [CLO-04] [20 marks]

Sketch the region enclosed by the given curves and find its area.

(a) $x = y^4, y = \sqrt{2 - x}\ , y = 0.$ (b) $y = 1/x\,,\quad y = x,\quad y = \frac{1}{4}x,\ x > 0.$

Question #5: [CLO-02] [10 marks]

Car A is traveling west at 50 mi/h and car B is traveling north at 60 mi/h. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mi and car B is 0.4 mi from the intersection?

Question #6: [CLO-04] [10 marks]. Evaluate the integral.

$$\int x^2\sqrt{(3 + 2x - x^2)}\,dx$$